

Abdulla Ansari

+36 70 576 4078 | abdulla.s.ansari247@gmail.com | abdulla-ansari.vercel.app | [abdulla-ansari247](https://www.instagram.com/abdulla-ansari247) | [abdulla-ansari-1247e](https://www.linkedin.com/in/abdulla-ansari-1247e)

Summary

Computational physics undergraduate at the University of Debrecen, graduating in January 2027 and **available to work in the Netherlands from 1 March**. I build simulations in C++ and Python: a year-long GEANT4 radiation-transport thesis running 1,000,000 gamma-ray primaries per configuration, and Monte Carlo reliability modelling in Python and SQL. Five years of self-sourced freelance software development alongside the degree keeps the engineering side sharp. Methodical and detail-oriented, and just as comfortable explaining the work to people who do not code. C2 English.

Education

Bachelor of Science - B.Sc Physics

UNIVERSITY OF DEBRECEN (DEBRECENI EGYETEM)

Debrecen, Hungary

Sept. 2023 - Jan. 2027

- Coursework across classical and quantum mechanics, electrodynamics, nuclear physics, and computational physics.

Experience

Battery Cell Quality & Sorting Operative

CATL (CONTEMPORARY AMPEREX TECHNOLOGY)

Debrecen, Hungary

Jun. 2026 - Present

- Work the production line at one of the world's largest lithium-ion battery manufacturers, assessing cell quality and sorting to tight tolerances.
- Full-time shift work taken on to fund my studies through thesis completion, in a precision-driven environment where consistency and attention to detail directly determine downstream product reliability.

Private Tutor

INDEPENDENT

Remote

Sept. 2020 - Jul. 2026

- Built a tutoring practice from scratch: my own students, my own rates, grown almost entirely by referral.
- Adapted my delivery to each student and to parents as the paying clients, holding attention over long-running one-to-one sessions.

Programming Developer

INDEPENDENT

Remote

Jan. 2021 - Present

- Deliver self-sourced client briefs end to end — scoping, agreeing the work, building, and following up — across Python, SQL, JavaScript (Node.js), Java, and HTML.
- Built an automated participant-management system for a client, handling registration and record-keeping for large participant groups across both the backend data-processing logic and the interactive web front-end.
- Build web tools and front-ends in Python, HTML, and CSS, including my portfolio site at abdulla-ansari.vercel.app — vanilla JS, persistent light/dark theming, and full EN/HU localisation via a custom i18n layer.

Relationship Manager

BELLAVISTA REAL ESTATE

Dubai, UAE

Feb. 2023 - Jul. 2023

- Acted as the face of the agency — opening conversations with new prospects, building trust quickly with buyers, sellers, and partner agencies, and negotiating between parties with competing interests in a fast-paced, target-driven market.

Projects

Optimization Of X-Ray Imaging Using GEANT4 For Enhanced Field-Of-View

BACHELOR THESIS

Debrecen, Hungary

Dec. 2025 - Jan. 2027 (expected)

- Designed and optimized a multi-source X-ray imaging configuration using GEANT4 simulations, running 1,000,000 gamma-ray primaries per configuration, to improve depth perception and structural resolution in conventional radiographic imaging systems.
- Developed computational models to simulate angled radiation sources and particle interactions, analyzing image reconstruction performance for detecting obscured structures within dense materials such as bone.
- Evaluated system performance through statistical analysis of imaging outputs, optimizing source geometry and beam orientation to enhance contrast, spatial resolution, and structural visibility compared to traditional single-source X-ray methods.

Offshore Wind Turbine Reliability Simulation

INDEPENDENT PROJECT

Dubai, UAE

May 2022 - Jul. 2022

- Built a Monte Carlo simulation in Python and SQL modelling failure rates and reliability behaviour of offshore wind turbine components, applying Poisson distribution methods to predict maintenance intervals and validating outputs against theoretical benchmarks.

2D AI Racer

INDEPENDENT PROJECT

In progress

- Top-down 2D racing game in Python with Pygame: vector-based car physics (acceleration, drag, trigonometric steering) with pixel-mask borders as groundwork for a self-driving agent that learns the racing line.

Skills

Programming	Python, C++, C, SQL, JavaScript (Node.js), Java, HTML, CSS
Tools & Frameworks	GEANT4, Linux, Git/GitHub, FastAPI, Pygame, Matplotlib, SciLab, Houdini
Domain Knowledge	Numerical modelling, statistical modelling, data analysis, Monte Carlo methods, simulation development
Languages	English (C2), Hindi (B1), Italian (A2), German (A1)